Honors Calculus 1610

Lecture 11: Extrema, the Mean Value Theorem, and the FTC

October 19, 2026

Today

- Where extrema can be: interior extrema are critical points.
- Rolle's theorem and the mean value theorem.
- What the MVT buys: the shape of a graph from the sign of f'.
- And then the payoff: the fundamental theorem of calculus.

Local extrema

Definition

Let $f : (a, b) \to \mathbb{R}$ and $c \in (a, b)$. We say c is a local maximum for f if there is $h > 0$ such that

$$f(c) \geq f(x) \qquad \text{for all } x \in (c - h, c + h) \cap (a, b),$$

and a local minimum if instead $f(c) \leq f(x)$ there.

Local: we only compare $f(c)$ with nearby values, not with all of them.

Interior extrema

Proposition

Suppose $f : (a, b) \to \mathbb{R}$ has a local maximum at $c \in (a, b)$ and that $f'(c)$ exists. Then $f'(c) = 0$.

The proof uses the same device as the chain rule. Set

$$q(x) = \begin{cases} \dfrac{f(x) - f(c)}{x - c}, & x \neq c, \\ f'(c), & x = c, \end{cases}$$

so that $\lim_{x \to c} q(x) = f'(c) = q(c)$: that is, q is continuous at c. This is just a restatement of what it means for $f'(c)$ to exist.

Proof

Suppose $f'(c) \neq 0$, say $q(c) = f'(c) > 0$.

By the sign preservation lemma — continuity at c with $\epsilon = |q(c)|$ — there is $h > 0$ with

$$q(x) > 0 \qquad \text{for all } x \in (c - h, c + h).$$

Pick $d \in (c, c + h)$. Then

$$\frac{f(d) - f(c)}{d - c} = q(d) > 0, \qquad d - c > 0,$$

so $f(d) > f(c)$, and c is not a maximum. If instead $f'(c) < 0$, take $d \in (c - h, c)$: now $d - c < 0$ and $q(d) < 0$, so again $f(d) > f(c)$. $\square$

Three cautions

- Interior matters. On $[0,1]$, $f(x) = x$ has a maximum at 1 and $f'(1) = 1 \neq 0$. The endpoint is not interior.
- $f'(c)$ must exist. $f(x) = -|x|$ has a maximum at 0 and no derivative there.
- The converse is false. $f(x) = x^3$ on $(-1,1)$ has $f'(0) = 0$ and no extremum at 0.

Rolle's theorem

Theorem

If f is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then $f'(c) = 0$ for some $c \in (a, b)$.

Note the asymmetry: continuity on the closed interval, differentiability only on the open one. The first is what the extreme value theorem needs.

Proof of Rolle

By the extreme value theorem f attains a minimum value m and a maximum value M on $[a, b]$.

- If either is attained at an interior point c, the proposition gives $f'(c) = 0$ and we are done.
- Otherwise both are attained only at the endpoints. Since $f(a) = f(b)$, that forces $m = M$, so f is constant — and then $f'(c) = 0$ for every $c \in (a, b)$.

□

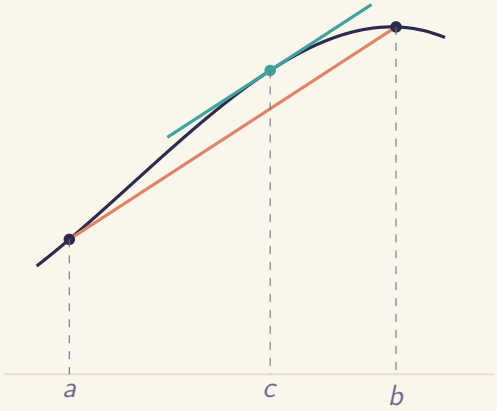
The mean value theorem

Theorem

If f is continuous on $[a, b]$ and differentiable on (a, b), then there is $c \in (a, b)$ with

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

The picture: some tangent is parallel to the chord



Proof of the mean value theorem

Let $m = \dfrac{f(b) - f(a)}{b - a}$ be the slope of the chord, and set

$$g(x) = f(x) - m\,x.$$

Then g is continuous on $[a, b]$ and differentiable on (a, b), and

$$g(b) - g(a) = f(b) - f(a) - m(b - a) = 0,$$

so $g(a) = g(b)$.

By Rolle there is $c \in (a, b)$ with $g'(c) = 0$, that is $f'(c) - m = 0$. $\square$

Graphing rules

Proposition

Let f be continuous on $[a, b]$ and differentiable on (a, b).

1. If $f'(x) > 0$ for all $x \in (a, b)$, then f is strictly increasing on $[a, b]$.
2. If $f'(x) < 0$ throughout, then f is strictly decreasing.
3. If $f'(x) = 0$ throughout, then f is constant.

One application of the MVT does all three

Take $x_1 < x_2$ in $[a, b]$. Then f is continuous on $[x_1, x_2]$ and differentiable on (x_1, x_2), so the MVT gives some $c \in (x_1, x_2)$ with

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = f'(c), \qquad \text{i.e.} \qquad f(x_2) - f(x_1) = f'(c)\,(x_2 - x_1).$$

The factor $x_2 - x_1$ is positive, so the sign of $f(x_2) - f(x_1)$ is exactly the sign of $f'(c)$. All three statements follow at once. $\square$

What this replaces

In Lecture 8 we showed x^n is increasing on $\mathbb{R}^+$ by manipulating inequalities, and on Homework 5 you showed a sum of increasing functions is increasing the same way.

All of that is now a single line: differentiate and look at the sign.

This is the first place where the derivative earns its keep as a tool rather than an object of study.

Critical points

Definition

If $f : (a, b) \to \mathbb{R}$ is differentiable, we say $c \in (a, b)$ is a critical point if $f'(c) = 0$.

So the proposition says: a local extremum is a critical point.

The converse fails — $f(x) = x^3$ on $(-1, 1)$ has $f'(0) = 0$ and no extremum. So critical points are candidates, and we need a way to sort them.

The first derivative test

Let c be a critical point, and suppose there is an interval $(c-h, c+h) \subseteq (a, b)$ on which

$$f \text{ is increasing on } (c-h, c) \quad \text{and} \quad f \text{ is decreasing on } (c, c+h).$$

Then f has a local maximum at c. Swap the two words and you get a local minimum.

More generally: a critical point is a local extremum as soon as f' changes sign there.

The second derivative test

Proposition

Let $f : (a, b) \to \mathbb{R}$ be differentiable, let $c \in (a, b)$ be a critical point, and suppose $f''(c)$ exists and $f''(c) \neq 0$. Then c is a local extremum — a minimum if $f''(c) > 0$, a maximum if $f''(c) < 0$.

We want to conclude that f' changes sign at c, and then quote the first derivative test.

Proof of the second derivative test

Say $f''(c) > 0$. Apply the same trick a third time, now to f':

$$Q(x) = \frac{f'(x) - f'(c)}{x - c} \ (x \neq c), \qquad Q(c) = f''(c),$$

which is continuous at c because $f''(c)$ exists.

Sign preservation gives $h > 0$ with $Q(x) > 0$ on $(c - h, c + h)$. But $f'(c) = 0$, so there $Q(x) = \dfrac{f'(x)}{x - c}$, and hence

$$f'(x) < 0 \text{ on } (c - h, c), \qquad f'(x) > 0 \text{ on } (c, c + h).$$

By the graphing rules f decreases then increases: a local minimum. $\square$

A remark on that proof

Applying sign preservation to Q instead reaches the same conclusion from the hypothesis as stated, and it reuses the device we had already used twice today.

The fundamental theorem

The fundamental theorem of calculus, I

Theorem

Let $f : [a, b] \to \mathbb{R}$, and suppose that for each $x \in [a, b]$ the integral

$$F(x) = \int_a^x f(t)\, dt$$

exists. If f is continuous at c, then F is differentiable at c and

$$F'(c) = f(c).$$

Differentiation undoes integration.

The difference quotient

$$\frac{F(c+h)-F(c)}{h} = \frac{\displaystyle\int_a^{c+h} f(t)\,dt - \int_a^{c} f(t)\,dt}{h} = \frac{1}{h}\int_c^{c+h} f(t)\,dt,$$

using additivity of the integral.

We want this to tend to $f(c)$. The trick is to put $f(c)$ into the integrand and show the rest is small:

$$\frac{1}{h}\int_c^{c+h} \Big(f(c) + \big(f(t) - f(c)\big)\Big)\,dt.$$

The main term

Split the integral in two (legitimate once we know both pieces converge):

$$\frac{1}{h}\int_c^{c+h} f(c)\,dt \;+\; \frac{1}{h}\int_c^{c+h} (f(t)-f(c))\,dt.$$

The first integrand is a constant — a step function — so

$$\frac{1}{h}\int_c^{c+h} f(c)\,dt = \frac{1}{h}\cdot f(c)\cdot h = f(c),$$

with no limit needed at all.

The error term

Claim

$$\lim_{h\to 0}\frac{1}{h}\int_c^{c+h}(f(t)-f(c))\,dt = 0.$$

Let $\epsilon > 0$. By continuity of f at c there is $\delta > 0$ with $|t-c| < \delta \implies |f(t)-f(c)| < \epsilon$. So for $0 < |h| < \delta$ the integrand is bounded by ϵ on the whole interval of integration, and

$$\left|\frac{1}{h}\int_c^{c+h}(f(t)-f(c))\,dt\right| \le \frac{1}{|h|}\cdot\epsilon\cdot|h| = \epsilon.$$

Hence $F'(c) = f(c)$. $\square$

The zero derivative theorem

Theorem

If f is continuous on $[a, b]$, differentiable on (a, b), and $f'(x) = 0$ for all $x \in (a, b)$, then f is constant.

This is case 3 of the graphing rules, and it is the MVT again: for any $x_1 < x_2$, $f(x_2) - f(x_1) = f'(c)(x_2 - x_1) = 0$.

Corollary

If $F' = G'$ on an interval, then $F - G$ is constant: $F = G + C$.

Antiderivatives

Definition

P is an antiderivative (or *primitive*) of f on $[a, b]$ if $P' = f$ there.

By the corollary, any two antiderivatives of the same function differ by a constant — so "the" antiderivative is well defined up to one number.

FTC I says that a continuous f has one, namely $F(x) = \int_a^x f$.

The fundamental theorem of calculus, II

Theorem

Let f be continuous and let P be any antiderivative of f. Then for every c,

$$P(x) = P(c) + \int_c^x f(t)\,dt.$$

Proof. By FTC I the right-hand side has derivative $f(x)$; by assumption so does P. Two functions with the same derivative differ by a constant. Setting $x = c$ makes both sides equal to $P(c)$, since $\int_c^c f = 0$ — so the constant is 0. $\square$

The evaluation rule

Take $c = a$ and $x = b$:

$$\int_a^b f(t)\,dt = G(b) - G(a) = \Big[G(x)\Big]_a^b$$

for any antiderivative G of f.

This is the computational payoff of the whole course so far: to integrate f, do not go back to step functions and suprema — run the derivative rules backwards.

Next time

A change of subject: vectors.

Still owed, and we will come back to them:

- $|x|$ is continuous but not differentiable at 0: one-sided limits.
- Higher derivatives and notation.
- Trig identities and $\frac{\sin x}{x} \to 1$, which is what we need before differentiating sin and cos.